QP CODE **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) MODEL EXAMINATION QUESTION PAPER, JANUARY, 2026

IV SEMESTER

SIGNATURE COURSE -

MG4DSEZGYA00 GENOME TECHNOLOGY

(2024 YEAR ADMISSION ONWARDS)

Duration: 2 Hrs **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer all questions

Each question carries 1 mark

1. Which biosafety level is appropriate for handling moderate-risk agents that cause human disease? [Remember / CO 1]

   (a) BSL-1
   (b) BSL-2
   (c) BSL-3
   (d) BSL-4

2. The enzyme used to detach adherent cells from the culture vessel surface is: [Remember / CO 1]

   (a) Collagenase
   (b) Trypsin
   (c) Ligase
   (d) Polymerase

3. Which component involves the use of dideoxynucleotides (ddNTPs) as chain terminators? [Understand / CO 3]

   (a) PCR
   (b) Sanger Sequencing
   (c) Southern Blotting
   (d) Karyotyping

4. The specific region of DNA where the CRISPR-Cas9 system binds and cuts is determined by: [Understand / CO 2]

   (a) Guide RNA (gRNA)
   (b) Palindromic Repeats
   (c) Cas9 enzyme
   (d) Viral DNA

5. A technique used to detect variations in the number of copies of a particular gene is: [Remember / CO 2]
   (a) RFLP

(b) Array CGH
(c) Sanger Sequencing
(d) Western Blotting

6. Which format is most commonly used to represent nucleotide sequences in bioinformatics? [Remember / CO 4]

   (a) JPEG
   (b) FASTA
   (c) CSV
   (d) HTML

7. In Next-Generation Sequencing (NGS), the term "reads" refers to: [Understand / CO 3]

   (a) The primers used
   (b) Short fragments of sequenced DNA
   (c) The computer software
   (d) The reference genome

8. The study of the complete set of RNA transcripts produced by the genome represents: [Remember / CO 3]

   (a) Genomics
   (b) Proteomics
   (c) Transcriptomics
   (d) Metabolomics

9. Which tool is primarily used for finding Open Reading Frames (ORFs)? [Apply / CO 4]

   (a) BLAST
   (b) ORF Finder
   (c) Clustal Omega
   (d) PyMol

10. The biological database GenBank is maintained by: [Remember / CO 4]

    (a) EMBL
    (b) DDBJ
    (c) NCBI
    (d) UCSC[

[1x10 = 10]

Part B

Short Answer Questions
Answer any 14 questions
Each question carries 2 marks

11. Define "confluency" in the context of cell culture. [Remember / CO 1]

12. Distinguish between primary cell culture and cell lines. [Understand / CO 1]

13. List two essential equipment items required in a cell culture laboratory and state their use. [Remember / CO 1]

14. What is a Single Nucleotide Polymorphism (SNP)? [Remember / CO 2]

15. Explain the significance of Satellite DNA. [Understand / CO 2]

16. What is the principle behind Reverse Transcription PCR (RT-PCR)? [Understand / CO 2]

17. Define Spectral Karyotyping (SKY) [Remember / CO 2]

18. What is the role of VNTRs in DNA fingerprinting? [Apply / CO 2]

19. Differentiate between Whole Exome Sequencing (WES) and Whole Genome Sequencing. [Analyse / CO 3]

20. Briefly explain the "chain termination" method in Sanger sequencing. [Understand / CO 3]

21. What is a "Gene Panel" analysis? Give one example. [Apply / CO 3]

22. Define Metabolomics and its integration with Genomics. [Remember / CO 3]

23. What is Maxam-Gilbert sequencing? [Remember / CO 3]

24. Explain the purpose of GENSCAN in genome annotation. [Apply / CO 4]

25. List the practical steps to retrieve a nucleotide sequence in FASTA format from the GenBank database. *[Apply / **CO 5**]*

26. What is the significance of the "E-value" in a BLAST search result? [Evaluate / CO 4]

[2x14 = 28]

Part C

Short Essay Questions

Answer any 8 questions

Each question carries 4 marks

27. Describe the procedure for initiating a primary cell culture from tissue, including the trypsinization process. [Apply / CO 1]

28. Elaborate on the various types of cell culture contamination and the aseptic techniques used to prevent them. [Understand / CO 1]

29. Explain the CRISPR-Cas9 mechanism and its potential applications in genome editing. [Analyse / CO 2]

30. Discuss the steps involved in Southern Blotting and its application in gene analysis. [Understand / CO 2]

31. Compare and contrast Conventional PCR and Quantitative PCR (qPCR). [Analyse / CO 2]

32. Provide an overview of Next-Generation Sequencing (NGS) technologies and their advantages over Sanger sequencing. [Understand / CO 3]

33. Explain the integration of Transcriptomics, Proteomics, and Metabolomics (Multi-Omics) in understanding biological systems. [Analyse / CO 3]

34. Describe the scope of Bioinformatics in modern Genomics. [Understand / CO 4]

35. Explain the steps involved in a phylogenetic analysis using MEGA software. [Apply / CO 4]

36. Explain the step-by-step procedure of a Conventional PCR reaction as performed in a laboratory or virtual simulation [Apply / CO 5]

[4x8 = 32]